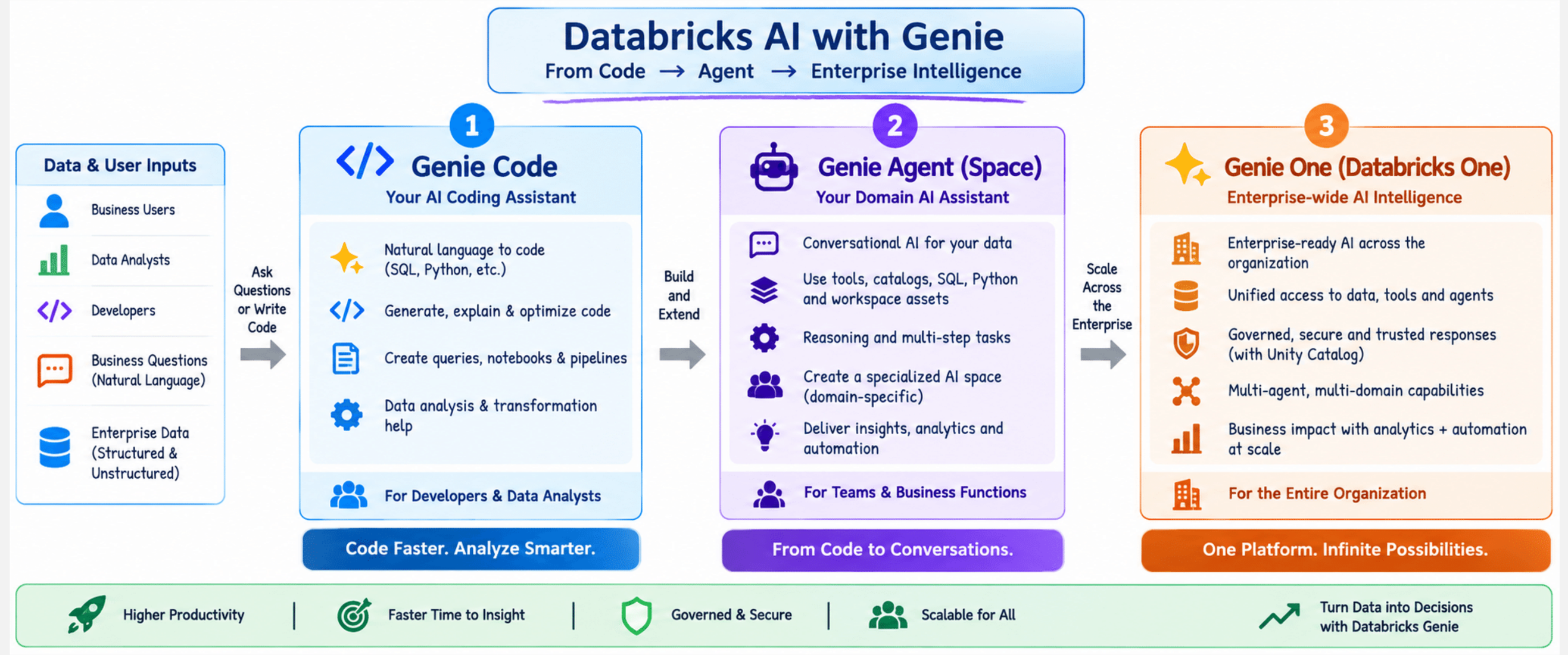


Databricks AI with Genie

From Code → Agent → Enterprise Intelligence

Data & User Inputs
- Business Users
- Data Analysts
- Developers
- Business Questions (Natural Language)
- Enterprise Data (Structured & Unstructured)

Ask Questions or Write Code

1 Genie Code
Your AI Coding Assistant
- Natural language to code (SQL, Python, etc.)
- Generate, explain & optimize code
- Create queries, notebooks & pipelines
- Data analysis & transformation help

For Developers & Data Analysts

Code Faster. Analyze Smarter.

Build and Extend

2 Genie Agent (Space)
Your Domain AI Assistant
- Conversational AI for your data
- Use tools, catalogs, SQL, Python and workspace assets
- Reasoning and multi-step tasks
- Create a specialized AI space (domain-specific)
- Deliver insights, analytics and automation

For Teams & Business Functions

From Code to Conversations.

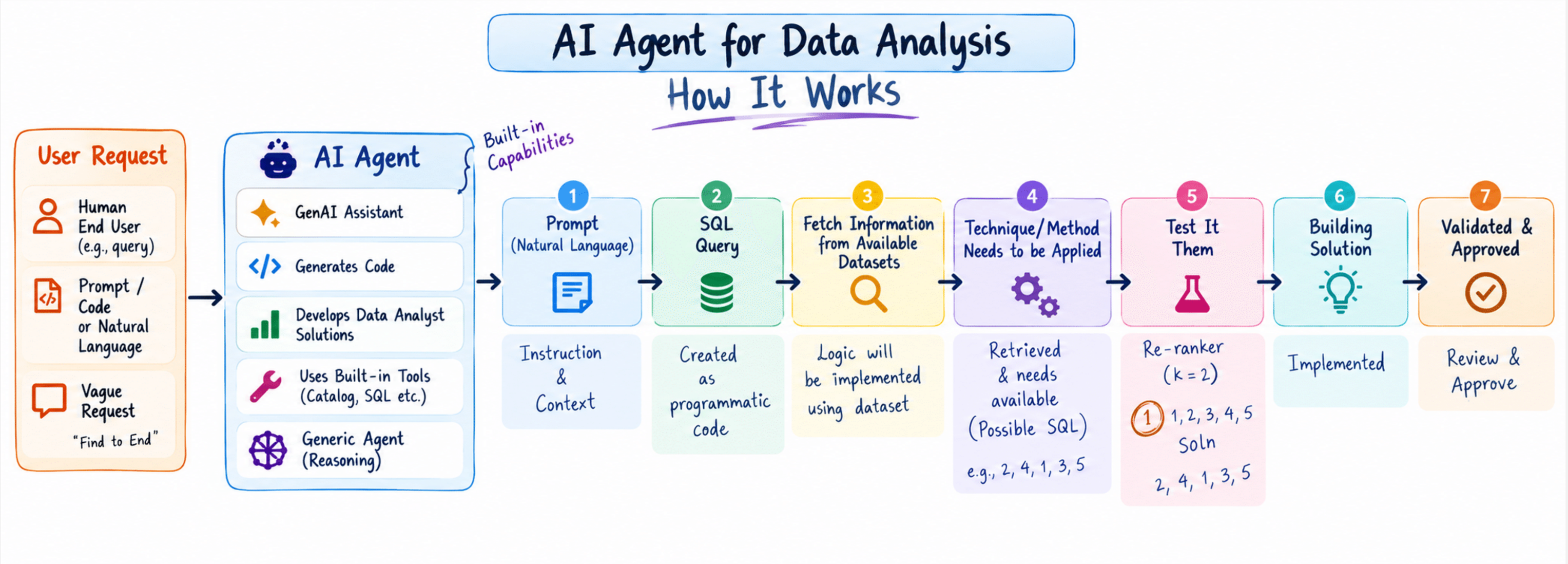
Scale Across the Enterprise

3 Genie One (Databricks One)
Enterprise-wide AI Intelligence
- Enterprise-ready AI across the organization
- Unified access to data, tools and agents
- Governed, secure and trusted responses (with Unity Catalog)
- Multi-agent, multi-domain capabilities
- Business impact with analytics + automation at scale

For the Entire Organization

One Platform. Infinite Possibilities.

Higher Productivity | Faster Time to Insight | Governed & Secure | Scalable for All | Turn Data into Decisions with Databricks Genie



AI Agent for Data Analysis

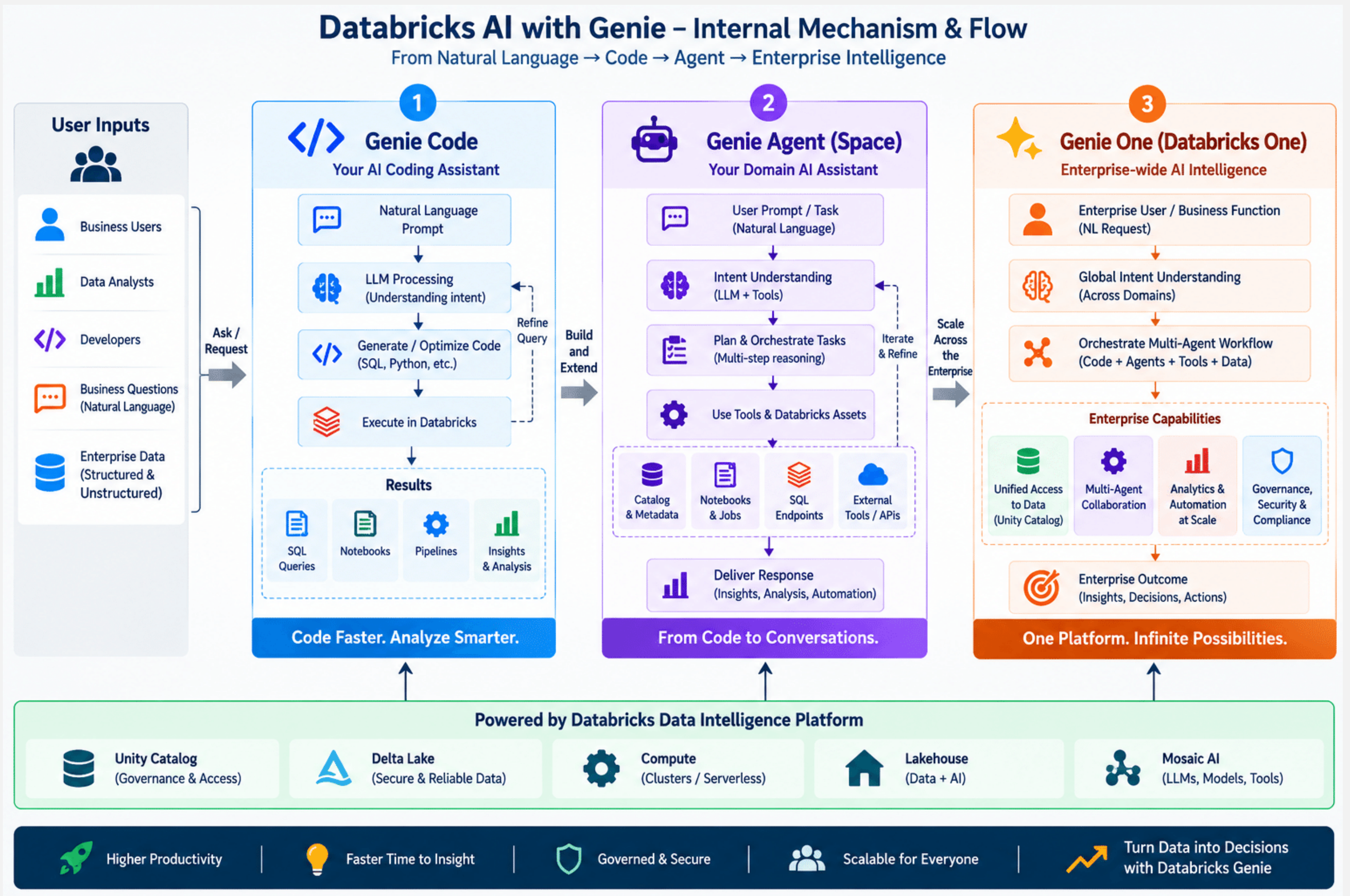
How It Works

User Request
- Human End User (e.g., query)
- Prompt / Code or Natural Language
- Vague Request "Find to End"

AI Agent
- GenAI Assistant
- Generates Code
- Develops Data Analyst Solutions
- Uses Built-in Tools (Catalog, SQL etc.)
- Generic Agent (Reasoning)

Built-in Capabilities

1. Prompt (Natural Language) — Instruction & Context
2. SQL Query — Created as programmatic code
3. Fetch Information from Available Datasets — Logic will be implemented using dataset
4. Technique/Method Needs to be Applied — Retrieved & needs available (Possible SQL) e.g., 2, 4, 1, 3, 5
5. Test It Them — Re-ranker (k = 2) ① 1, 2, 3, 4, 5 Soln 2, 4, 1, 3, 5
6. Building Solution — Implemented
7. Validated & Approved — Review & Approve



Databricks AI with Genie – Internal Mechanism & Flow

From Natural Language → Code → Agent → Enterprise Intelligence

User Inputs
- Business Users
- Data Analysts
- Developers
- Business Questions (Natural Language)
- Enterprise Data (Structured & Unstructured)

Ask / Request

1 Genie Code
Your AI Coding Assistant

Natural Language Prompt → LLM Processing (Understanding intent) → Generate / Optimize Code (SQL, Python, etc.) → Execute in Databricks (Refine Query)

Results: SQL Queries, Notebooks, Pipelines, Insights & Analysis

Code Faster. Analyze Smarter.

Build and Extend

2 Genie Agent (Space)
Your Domain AI Assistant

User Prompt / Task (Natural Language) → Intent Understanding (LLM + Tools) → Plan & Orchestrate Tasks (Multi-step reasoning) → Use Tools & Databricks Assets (Iterate & Refine)

Catalog & Metadata, Notebooks & Jobs, SQL Endpoints, External Tools / APIs

Deliver Response (Insights, Analysis, Automation)

From Code to Conversations.

Scale Across the Enterprise

3 Genie One (Databricks One)
Enterprise-wide AI Intelligence

Enterprise User / Business Function (NL Request) → Global Intent Understanding (Across Domains) → Orchestrate Multi-Agent Workflow (Code + Agents + Tools + Data)

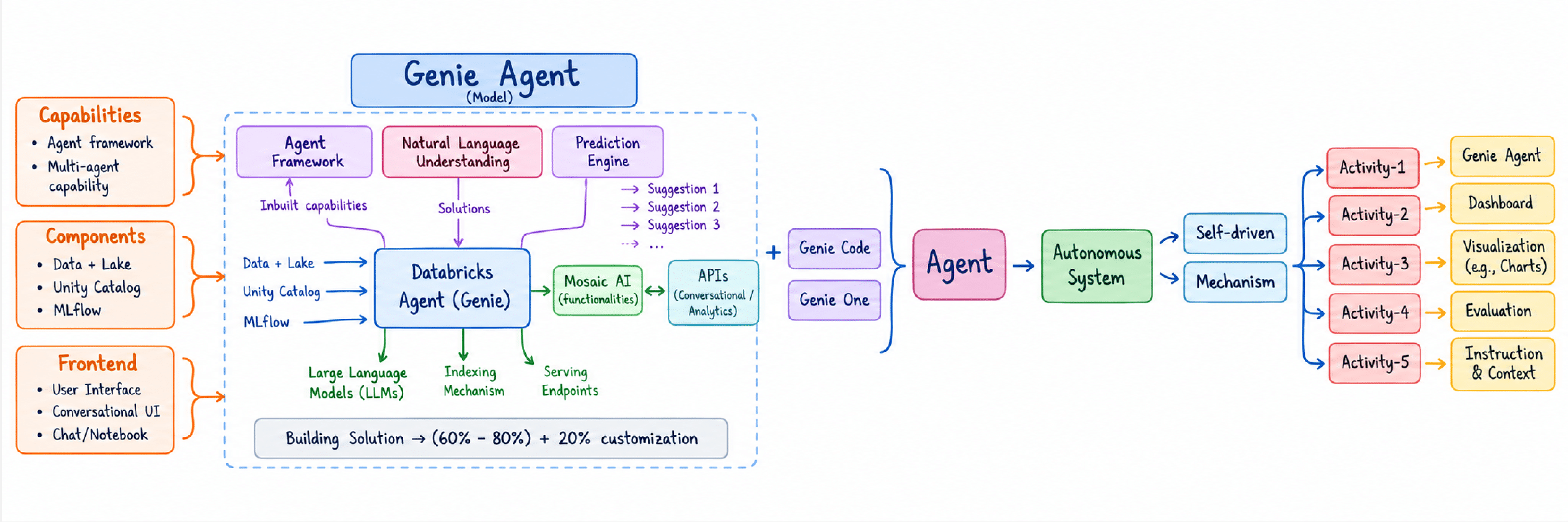
Enterprise Capabilities: Unified Access to Data (Unity Catalog), Multi-Agent Collaboration, Analytics & Automation at Scale, Governance, Security & Compliance

Enterprise Outcome (Insights, Decisions, Actions)

One Platform. Infinite Possibilities.

Powered by Databricks Data Intelligence Platform
- Unity Catalog (Governance & Access)
- Delta Lake (Secure & Reliable Data)
- Compute (Clusters / Serverless)
- Lakehouse (Data + AI)
- Mosaic AI (LLMs, Models, Tools)

Higher Productivity | Faster Time to Insight | Governed & Secure | Scalable for Everyone | Turn Data into Decisions with Databricks Genie



Genie Agent

(Model)

Capabilities
- Agent framework
- Multi-agent capability

Components
- Data + Lake
- Unity Catalog
- MLflow

Frontend
- User Interface
- Conversational UI
- Chat/Notebook

Agent Framework | Natural Language Understanding | Prediction Engine

Inbuilt capabilities · Solutions · → Suggestion 1 → Suggestion 2 → Suggestion 3 ⇢ …

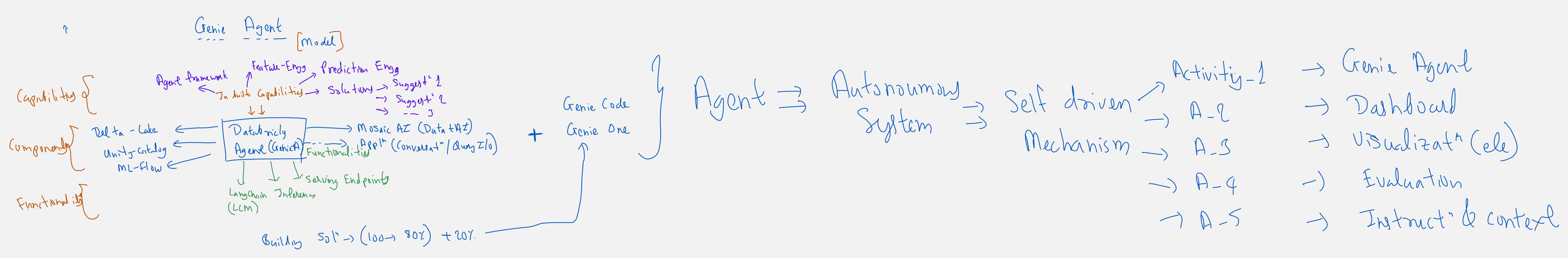
Data + Lake → Unity Catalog → MLflow → Databricks Agent (Genie) → Mosaic AI (functionalities) ↔ APIs (Conversational / Analytics)

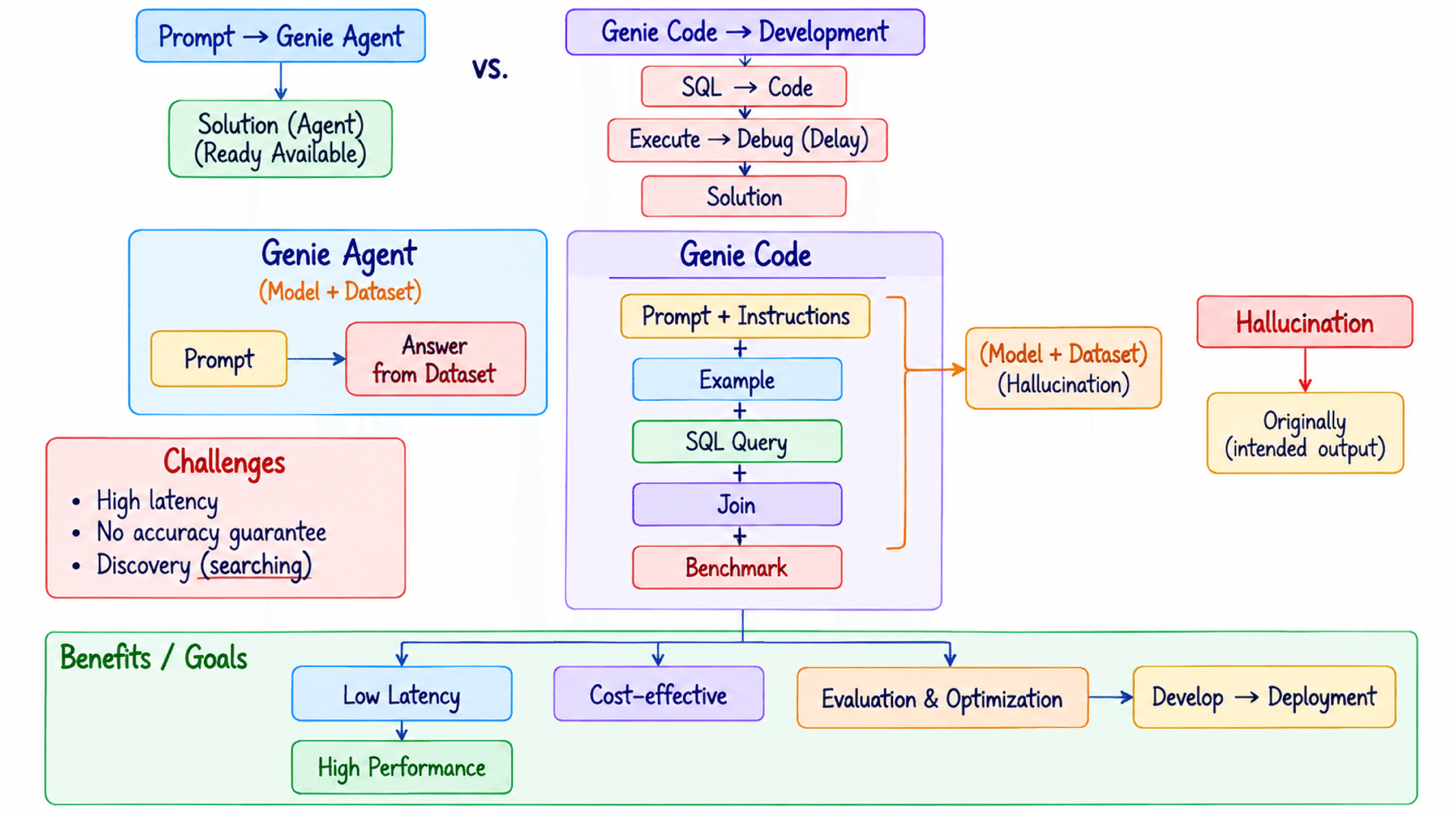
Large Language Models (LLMs) · Indexing Mechanism · Serving Endpoints

Building Solution → (60% – 80%) + 20% customization

+ Genie Code, Genie One → Agent → Autonomous System → Self-driven, Mechanism

- Activity-1 → Genie Agent
- Activity-2 → Dashboard
- Activity-3 → Visualization (e.g., Charts)
- Activity-4 → Evaluation
- Activity-5 → Instruction & Context





Prompt → Genie Agent → Solution (Agent) (Ready Available)

vs.

Genie Code → Development → SQL → Code → Execute → Debug (Delay) → Solution

Genie Agent
(Model + Dataset)
Prompt → Answer from Dataset

Challenges
- High latency
- No accuracy guarantee
- Discovery (searching)

Genie Code
Prompt + Instructions + Example + SQL Query + Join + Benchmark → (Model + Dataset) (Hallucination)

Hallucination → Originally (intended output)

Benefits / Goals
- Low Latency → High Performance
- Cost-effective
- Evaluation & Optimization → Develop → Deployment